Philippine Physics Olympiad (PPO) National Finals 2026

Sample Problems

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Q1 (Morning Medley, 16 pts total)

This problem consists of five (5) independent sub-problems. The five parts can be solved in any order.

1A) Spring with Mass

We usually ignore the kinetic energy of the moving coils of a spring, but let's try to get a reasonable approximation to this. Consider a spring of mass M and spring constant k that has one end fixed and the other end moving with speed v. Assume that the speed of the points along the length of the springs varies linearly with distance from the fixed end. Assume also that the mass M of the spring is distributed uniformly along the length of the spring.

1.1 Determine the kinetic energy of the spring in terms of M and v. **[2 pts]**

Suppose that a block of mass $2M$ is attached to the free end of the spring and allowed to oscillate on a horizontal, frictionless surface.

1.2 Determine the angular frequency ω of the block. **[2 pts]**

1B) Leaky Capacitor

A parallel plate capacitor is initially in a vacuum. The space between the plates is then completely filled with a slightly conducting dielectric material with a dielectric constant $\kappa = 30$ and a finite electrical resistivity ρ. The capacitor is charged with an initial voltage and then completely isolated from any external circuit.

1.3 Determine the time $t_{1/3}$ it takes for the electrical energy stored in the capacitor to drop to exactly $1/3$ of its initial value. Express your answer in terms of ρ and the vacuum permittivity ϵ_0. **[3 pts]**

1C) Falling Speaker

A Bluetooth speaker emitting a continuous sound of constant frequency f_0 is dropped from rest at a height H above a smartphone resting on the ground. The smartphone continuously records the frequency of the sound it receives. The acceleration due to gravity is g, and the speed of sound in air is u Assume air resistance is negligible and that the speaker does not rotate. Just before impact, the smartphone records a maximum frequency of $5f_0/4$.

1.4 Express the initial height H strictly in terms of g and u. **[3 pts]**

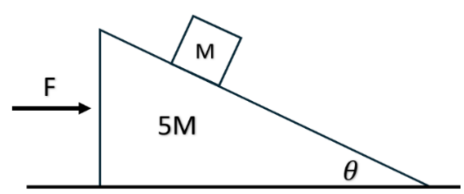
1D) Non-Standard Expansion

One mole of an ideal monatomic gas undergoes a quasi-static thermodynamic process where its pressure varies linearly with the volume. The gas expands from an initial state with temperature T_0 to a final state with a volume twice the original.

1.5 Determine the total heat added to the gas during this process. **[3 pts]**

1E) Movable Wedge

A wedge of mass 5M and inclination angle θ rests on a frictionless horizontal tabletop. A block of mass M is placed on the wedge, and a horizontal force F is applied to the wedge, as shown in the figure.



1.6 Determine the value of F such that the block remains at a constant height. **[3 pts]**

Q2 (Scattering, 12 pts total)

In 1911, the Rutherford scattering experiment revolutionized our understanding of the atom by demonstrating the existence of a dense, positively charged nucleus. In this problem, we will explore the elegant symmetries of two-body Coulomb scattering.

Consider an alpha particle of mass m and positive charge q, and a target nucleus of positive charge Q and mass $M = \eta m$, where η is a dimensionless mass ratio. Let the target nucleus be a Sulfur-32 nucleus, so that $\eta = 8$.

To simplify your analysis, we define the characteristic scattering length of the system as:

$$r_0 = \frac{1}{4\pi\epsilon_0}\frac{qQ}{mv_0^2}$$

where v_0 is a reference speed.

For Parts 2A-2D, assume all speeds are strictly non-relativistic. For all problems, neglect any energy lost to electromagnetic radiation during the acceleration of the charges, and assume that gravitational interactions are negligible.

2A) Head-On Trajectory

Assume the target nucleus is fixed rigidly in space. The alpha particle is fired from infinity with an initial speed v_0 directly at the nucleus (impact parameter $b = 0$).

2.1 Express the distance of closest approach d_A in terms of r_0.**[1 pt]**

2B) Grazing Trajectory

Assume the nucleus is still fixed rigidly in space. The alpha particle is again fired with initial speed v_0, but its trajectory is now offset by a perpendicular impact parameter b.

2.2 Calculate the required impact parameter such that the alpha particle's distance of closest approach is exactly $d_B = 4r_0$.**[2 pt]**

2C) Recoiling Target

Now assume the target nucleus is initially at rest but is completely free to move and recoil. The alpha particle is fired from infinity with speed v_0 and impact parameter b.

2.3 Find the new required impact parameter, still requiring that the distance of closest approach is $4r_0$.**[2 pt]**

2D) Galilean Invariance

Suppose the nucleus is also initially moving along the $+x$ axis with speed $v_2 = 0.5v_0$. The alpha particle is fired from behind it along the same axis in a head-on trajectory ($b = 0$).

2.4 To achieve that same target distance of closest approach $4r_0$, the alpha particle must be fired with a new initial speed v_1. Determine v_1 strictly in terms of v_0. **[3 pt]**

2E) Relativistic Recoil

Suppose that the classical alpha particle is replaced by a highly energetic gamma-ray photon of incident wavelength λ_0. The photon is fired at the same initially stationary target nucleus and undergoes a direct quantum backscatter (180° reflection). Let the Compton wavelength of the alpha particle be defined as $\lambda_C = h/mc$.

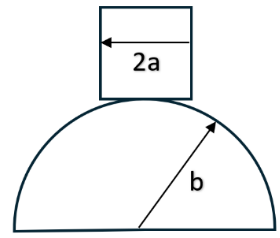
2.5. Determine the exact incident wavelength λ_0 required to cause the recoiling target nucleus to achieve a kinetic energy exactly equal to its own rest energy. Express your final answer strictly in terms of λ_C. **[4 pts]**

Q3 (Rocking Rigid Bodies, 15 pts total)

This problem investigates the dynamics of constrained mechanical systems to determine stability and characteristic motion.

3A) Cubical Block on a Sphere

A uniform cubical block of mass m and side length $2a$ is balanced in perfect equilibrium on the apex of a fixed, rough spherical boulder of radius b. Suppose the block is slightly perturbed such that it rocks without slipping. Let θ be the angle of the instantaneous contact point measured from the vertical.



3.1 Determine the potential energy function of the block in terms of $m, g, a, b,$ and θ. **[3 pt]**

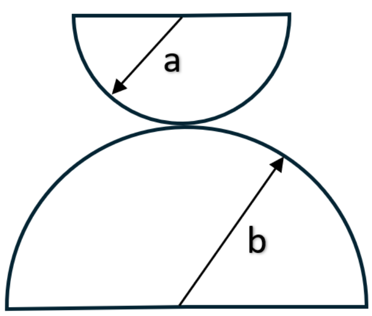
3.2 Determine the condition on a and b for which the system is in stable equilibrium. **[1 pts]**

When the stability condition is satisfied, a small displacement from the apex will cause the block to experience a restoring torque, resulting in periodic motion about its equilibrium state. Consider a specific configuration where $a = R$ and $b = 3R$.

3.3 Determine the period T_1 of small amplitude oscillations of the block. Express your answer in terms of R and g only. **[3 pts]**

3B) The Rocking Dome

Now, we replace the cubical block with a solid, homogeneous hemisphere of mass M and radius a. It rests in perfect equilibrium on the apex of the same fixed boulder of radius b. The curved faces of both hemispheres are in contact, and friction is sufficiently large to prevent any slipping.



3.4 Determine the distance from the instantaneous point of contact to the center of mass of the upper hemisphere when it is in the upright position. **[1 pt]**

Suppose that the upper hemisphere is slightly perturbed such that it rolls without slipping. The point of contact moves by a small angle θ along the surface of the lower hemisphere.

3.5 Express the total angle of rotation ψ of the upper hemisphere in the stationary frame in terms of $\theta, a,$ and b. **[2 pts]**

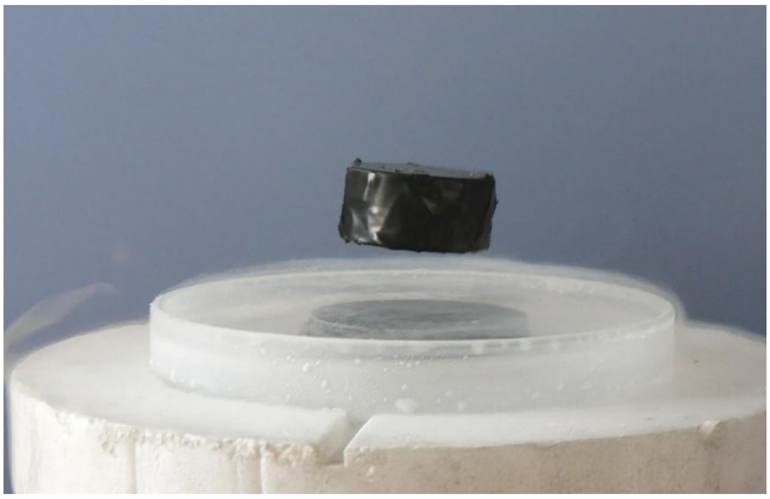
3.6 Determine the condition on a and b for which the system is in stable equilibrium. **[2 pts]**

Consider a specific configuration where $a = 2R$ and $b = 5R$.

3.7 Assuming that the stability condition is satisfied, determine the period of small amplitude oscillations T_2 of the upper hemisphere. **[3 pts]**

Q4 (Superconducting Levitation, 11 pts total)

A familiar demonstration of superconductivity is the levitation of a magnet over a piece of superconducting material. This phenomenon is governed by the Meissner effect, which is the complete expulsion of external magnetic fields from the interior of a material as it transitions into a superconducting state.



Source: https://www.britannica.com/science/Meissner-effect#/media/1/373488/152927

The magnetic field produced by a dipole $\vec{\mu}$ at a distance r along its primary axis is given by:

$$\vec{B} = \frac{\mu_0 \mu}{2\pi r^3}\hat{\jmath}$$

where μ_0 is the vacuum permeability (not to be confused with the magnetic moment μ!). The potential energy of a dipole in an external magnetic field is $U = -\vec{\mu} \cdot \vec{B}$.

4A) Magnetic Method of Images

Consider a small permanent magnet of mass m and a magnetic dipole moment $\vec{\mu} = \mu\hat{\jmath}$ that is constrained to move only along the vertical y-axis. It is placed a height d above an infinite, flat superconducting slab that occupies the entire half-space $y \leq 0$. Due to the perfect diamagnetism of the superconductor, the boundary condition requires that the normal component of the magnetic field vanishes exactly at the surface ($y = 0$). This boundary condition can be satisfied using the method of images.

4.1 Deduce the magnitude, direction, and vertical location of the image dipole $\vec{\mu}'$ that satisfies the appropriate boundary condition for the magnetic field at the surface, $y = 0$, given the real magnet is at an instantaneous height d. **[2 pts]**

4.2 Derive an expression for the magnitude of the upward magnetic force $\lvert F(d) \rvert$ acting on the real magnet as a function of its height d above the superconductor. **[2 pts]**

4B) Equilibrium State

Assume a uniform gravitational field g acts in the system and that air resistance is negligible.

4.3 Express the equilibrium levitation height d of the magnet strictly in terms of $m, \mu, g,$ and μ_0. **[2 pts]**

4C) Vertical Oscillation

Suppose the levitating magnet is given a small gentle vertical tap.

4.4 Determine the period of the resulting small vertical oscillations strictly in terms of d and g. **[5 pts]**

Q5 (Afternoon Medley, 16 pts total)

This problem consists of four (4) independent sub-problems. The four parts can be solved in any order.

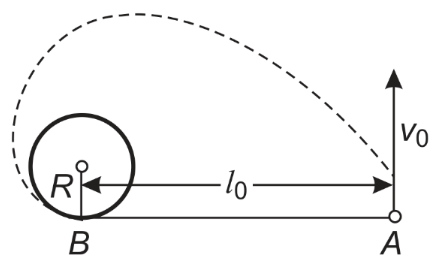
5A) Feynman Disk Paradox

A thin, non-conducting ring of mass m and radius R carries a total positive charge Q uniformly distributed along its circumference. The ring lies on a frictionless, horizontal non-conducting table and is free to rotate about its vertical central axis. Initially, the ring is at rest. A uniform magnetic field of magnitude B is applied perpendicular to the plane of the ring. However, the magnetic field is confined strictly to a central circular region of radius $R/2$, concentric with the ring. The magnetic field is then gradually reduced to zero.

5.1 Determine the angular speed of the ring after the magnetic field has completely vanished. Express your final answer in terms of $m, Q,$ and B.**[4 pts]**

5B) Involute Constraint

Consider a small disc of mass m placed on a horizontal surface. At the center of this surface is a fixed vertical cylinder of radius R. The disc is attached to the cylinder by a non-extensible, horizontal thread of initial length l_0. At $t = 0$, the thread is fully extended and tangent to the cylinder at point B. The disc is then imparted an initial velocity v_0 perpendicular to the thread, as shown in the figure. The surface is not frictionless. Instead, it exerts a kinetic friction force such that the speed of the disc decreases according to $v(r) = v_0\sqrt{r/l_0}$ where r is the remaining length of the free thread.



5.2 Determine the time t_{hit} required for the disc to strike the surface of the cylinder. **[4 pts]**

5C) Leaking Vertical Slit

A cubical tank of edge length L is initially full of water and rests on a horizontal surface. The tank has a vertical slit extending from the top to the bottom of one of its faces. The width of the slit is given by αL, where $\alpha \ll 1$. The top of the tank is open to the atmosphere, and the water exits the slit directly into the surrounding air. Assume the water behaves as an ideal, incompressible, and non-viscous fluid. Surface tension, edge effects, and the contraction of the exiting jet is also assumed to be negligible.

5.3 Determine the time $t_{3/4}$ it takes for the water surface to fall by a distance $3L/4$ from the top of the tank. Express your answer in terms of α, g and L. **[4 pts]**

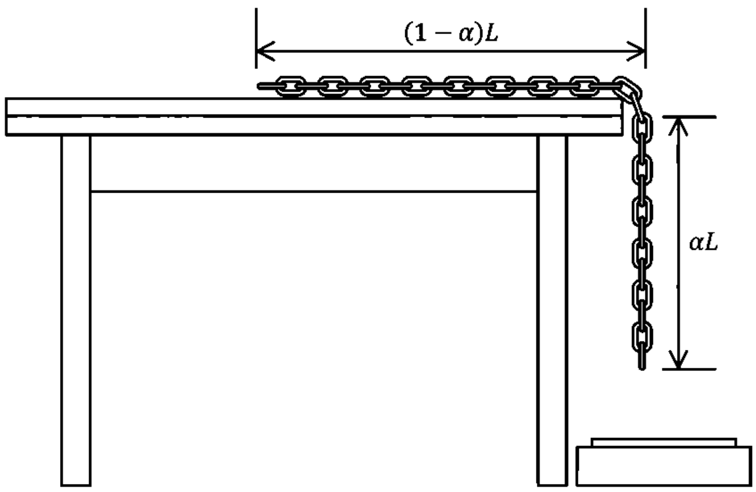
5D) Rotating Disk

A thin dielectric disk with radius R has a total charge Q distributed uniformly over its surface. The disk rotates n times per second about an axis perpendicular to the surface of the disk and passing through its center.

5.4 Find the magnetic field at the center of the disk in terms of $R, Q, n,$ and the vacuum permeability μ_0. **[4 pts]**

Q6 (The Falling Chain, 11 pts total)

In this problem, you will consider the mechanics of a sliding chain. Consider a uniform steel chain of total mass M and length L resting on a rough horizontal table. At $t = 0$, a fraction of its length αL (where $\alpha < 1$) hangs vertically over the edge of the table, and the chain is held at rest.



6A) The Tipping Point
Let the coefficient of static and kinetic friction between the chain and the table be $3/4$ and $1/2$, respectively.

6.1 Derive the general condition for α for which the chain will begin to slide from rest if released. **[2 pts]**

For simplicity, let $\alpha = 2/3$ for the rest of the problems below.

6.2 Determine the speed of the chain at the exact instant the last link leaves the table. **[2 pts]**

6B) The Sizzling Slide
As the chain slides off the table, friction generates thermal energy. For simplicity assume that the chain absorbs all the thermal energy generated by friction and that the heat distributes uniformly throughout the chain. Let c and γ be the specific heat and coefficient of linear thermal expansion of the chain, respectively.

6.3 Determine the fractional change in length of the chain at the instant the last link leaves the table. Express your answer in terms of $\gamma, g, L,$ and c. **[3 pts]**

6C) The Impact
Finally, after leaving the table, the chain falls vertically. Assume that the thermal expansion from Problem 2B is negligible for the kinematic calculations of its fall. The chain falls into the pan of the weighing scale located at a height L below the edge of the table and coils without bouncing.

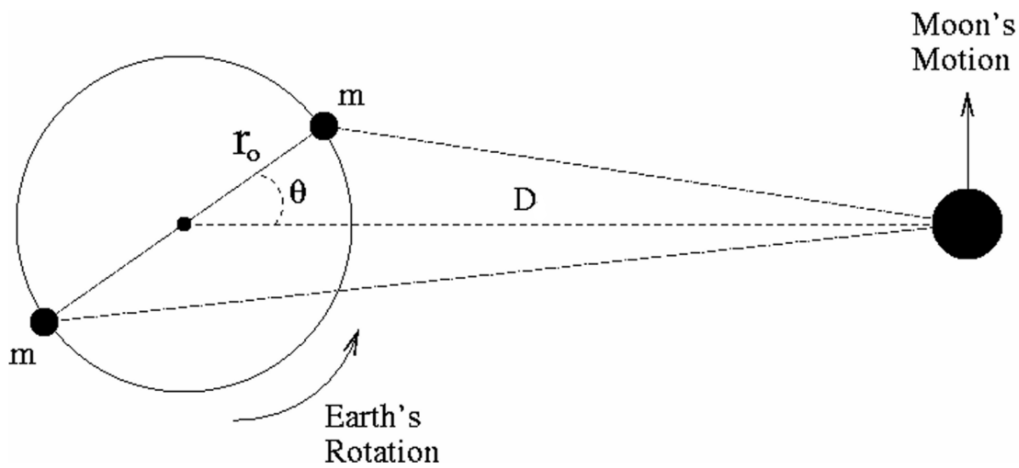
6.4 Determine the maximum reading recorded by the scale as a function of M and g. **[4 pts]**

Q7 (Tidal Torques, 10 pts total)

(**Source:** International Physics Olympiad 2009)

We can determine the Earth-Moon distance with great precision by bouncing laser beams off retroreflectors deposited on the Moon's surface during the Apollo missions and measuring the round-trip travel time of the light. Through these observations, it has been directly measured that the Earth-Moon distance is slowly increasing over time. This recession occurs because tidal torques are gradually transferring spin angular momentum from the Earth to the orbital angular momentum of the Moon.

In this problem, you will derive the basic parameters of this phenomenon using a simplified model. Assume that the Earth's tidal bulges can be approximated by two point masses, each of mass m, located on the surface of the Earth at a distance r_0 from the Earth's center. Let θ be the angle between the axis passing through the two bulges and the line joining the centers of the Earth and the Moon. Let the mass of the Moon be M_M, the mass of the Earth be M_E, and the distance between their centers be D. Assume the Moon is in a circular orbit and that $r_0 \ll D$.



7A) Tidal Forces and Torques

7.1 Determine the magnitude of the gravitational force exerted on the Moon by the farthest tidal bulge. Express your answer in terms of G, m, M_M, D, r_0, and θ. **[1 pts]**

7.2 Determine the magnitude of the net torque exerted on the Moon by the two tidal bulges with respect to the center of the Earth. Since $r_0 \ll D$, approximate your expression to the lowest significant order in r_0/D. **[2 pts]**

7B) Orbital Recession and Earth's Rotation

As a result of this torque, the Moon's orbital angular momentum increases, causing it to recede. Due to the conservation of angular momentum of the isolated Earth-Moon system, the Earth's rotation slows down as the Moon recedes.

7.3. Find the rate of increase of the Earth-Moon distance per year. Express your answer in terms of τ, G, M_E, M_M, and D. **[4 pts]**

7.4. Determine the rate at which the length of the day is currently increasing. **[3 pts]**

Q8 (Protostar Formation, 15 pts total)

(**Source:** International Physics Olympiad 2012)

This problem investigates the dynamical evolution of a star during its initial formation from an interstellar gas cloud. Consider a spherical cloud of mass M and initial radius r_0, composed of a sparse ideal gas with average molar mass μ and adiabatic index $\gamma > 4/3$. The cloud is initially at rest and in thermodynamic equilibrium with its surroundings at a constant temperature T_0. Assume that gravitational effects dominate thermal pressure, such that $GM\mu/r_0 \gg RT_0$, where G is the gravitational constant and R is the molar gas constant.

8A) Initial Stages

Neglect the change in the gravitational field strength at the position of a falling gas particle.

8.1 Estimate the time $t_{collapse}$ for the cloud's radius to shrink from r_0 to $r_1 = 0.95r_0$. **[3 pts]**

During much of the collapse, the gas is so transparent that any heat generated is immediately radiated away, i.e. the cloud stays in thermodynamic equilibrium with its surroundings. Assume that the gas density remains uniform throughout this phase.

8.2 Determine the factor N by which the internal pressure increases when the cloud radius shrinks to $r_2 = 0.5r_0$. **[2 pts]**

8B) Radiated Heat

At some specific radius $r_3 \ll r_0$, the gas becomes dense enough to be opaque to its own radiation.

8.3. Calculate the total amount of heat radiated away during the collapse from r_0 down to r_3. Express your answer in terms of G, M, R, μ, T_0, r_3, and r_0. **[3 pts]**

For even smaller radii $r < r_3$, heat loss due to radiation becomes negligible.

8.4. Determine the functional dependence of the temperature T of the spherical cloud on its radius r in this opaque phase. **[3 pts]**

8C) Final Radius

Eventually we cannot neglect the effect of the pressure on the dynamics of the gas and the collapse stops at $r = r_4$ where $r_4 \ll r_3$. However, the radiation loss can still be neglected, and the temperature is not yet high enough to ignite nuclear fusion. The pressure of such a protostar is not uniform anymore, but rough estimates can still be done.

8.5. Estimate the final radius r_4 and the respective temperature T_4. **[4 pts]**